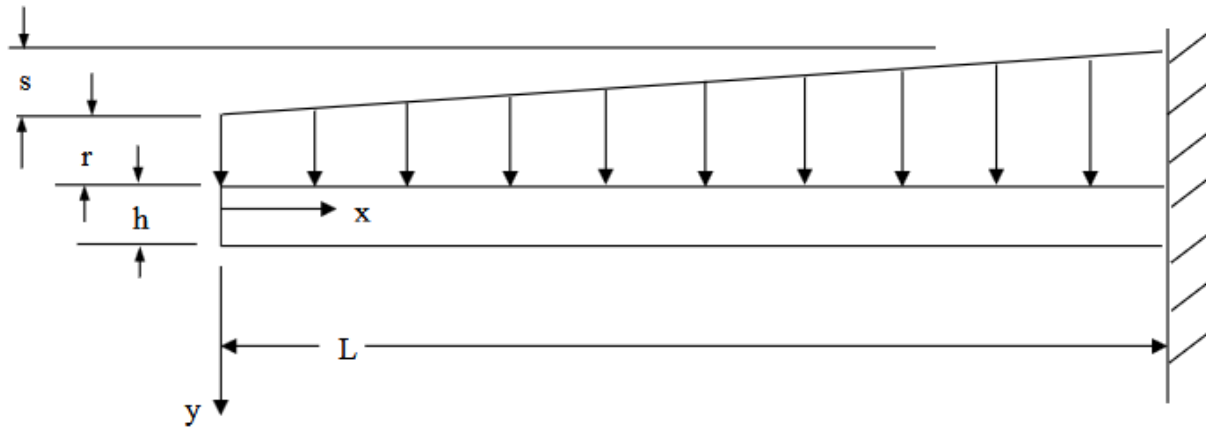


Samuel Roth
Date: 4/17/2026

Spring 2026: AEE 300
Design and Analysis
of Aerospace Structures

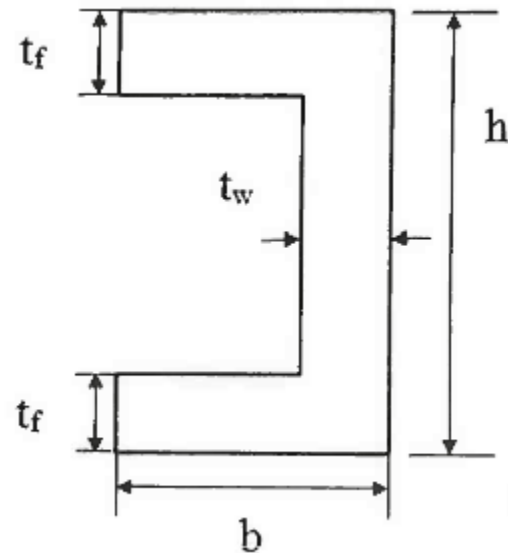
Design Project 2 Report

Design of an Aircraft Main Wing Spar



DESIGN CONSTRAINTS

$L = 120$ in
 $4.0 < h < 8.0$ in
 $3.0 < b < 6.0$ in
 $t_f > 0.135$ in
 $t_w > 0.135$ in
 $0.5 < t_w/t_f < 2.0$
 $b_f/b_w < 1.0$
 # Flights to Failure (F) > 10000
 Scatter Factor (SF) = 4.0
 FS (strength) > 1.500
 FS (yield) > 1.250



Constraint check using root design:

$h = 7.750$ in --> 1 $b = 5.250$ in --> 1
 $t_f = 0.336$ in --> 1 $t_w = 0.537$ in --> 1
 $t_w/t_f = 1.598$ --> 1 $b_f/b_w = 0.333$ --> 1

GOVERNING STATIC LOADING CASE

Governing load case: Maneuver type 2 MAX

$r = 60.000$ lb/in

$s = 50.000$ lb/in

(Limit) Root bending moment = 552.000 kip-in

Moment equation used: $M(x) = r \cdot x^2/2 + s \cdot x^3/(6 \cdot L)$

This case drives the static sizing used for the spanwise design check.

Section 1: x = 0 in

Requirements

(Limit) Bending Moment (in-lb) = 0.000

Design Geometry

Width h (in) = 4.000

Depth, b (in) = 3.000

Flange thickness tf (in) = 0.135

Web thickness tw (in) = 0.135

Failure Predictions

Critical stress - Crippling (psi) = 32181.433

Critical stress - Local Buckling (psi) =
51747.062

Analysis Information

Cross-sectional area (in²) = 1.314Moment of Inertia Iz (in⁴) = 3.610

Local Buckling Coefficient, kw = 4.000

(Limit) Max. Bending Stress (psi) = 0.000

(Limit) Avg. Corner Stress (psi) = 0.000

(Limit) Avg. Flange Stress (psi) = -0.000

Factors of Safety

Ultimate Strength = Inf

Yielding = Inf

Crippling = Inf

Local Buckling = Inf

Table 1A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 1

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	-0.000	-0.000	NaN	-0.000	0.000	Inf
2	-0.000	-0.000	NaN	-0.000	0.000	Inf
3	-0.000	-0.000	NaN	-0.000	0.000	Inf
4	-0.000	-0.000	NaN	-0.000	0.000	Inf
5	-0.000	-0.000	NaN	-0.000	0.000	Inf
6	-0.000	-0.000	NaN	-0.000	0.000	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Table 1B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 1

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	0.000	0.000	NaN	0.000	0.000	Inf
2	0.000	0.000	NaN	0.000	0.000	Inf
3	0.000	0.000	NaN	0.000	0.000	Inf
4	0.000	0.000	NaN	0.000	0.000	Inf
5	0.000	0.000	NaN	0.000	0.000	Inf
6	0.000	0.000	NaN	0.000	0.000	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Section 2: $x = 12$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 4440.000

Design Geometry

Width h (in) = 5.429Depth, b (in) = 3.674Flange thickness t_f (in) = 0.172Web thickness t_w (in) = 0.209

Failure Predictions

Critical stress - Crippling (psi) = 35170.746

Critical stress - Local Buckling (psi) =
66942.152

Analysis Information

Cross-sectional area (in²) = 2.329Moment of Inertia I_z (in⁴) = 11.037Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 1092.069

(Limit) Avg. Corner Stress (psi) = 1022.810

(Limit) Avg. Flange Stress (psi) = 1057.440

Factors of Safety

Ultimate Strength = 74.171

Yielding = 65.014

Crippling = 32.206

Local Buckling = 61.298

Table 2A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 2

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-706.598	-1503.514	2.128	-0.707	0.956	Inf
2	-648.158	-1583.205	2.443	-0.648	0.978	Inf
3	-544.559	-1638.104	3.008	-0.545	0.963	Inf
4	-941.423	-1412.134	1.500	-0.941	0.941	Inf
5	-706.598	-1503.514	2.128	-0.707	0.956	Inf
6	1089.118	-134.590	-0.124	1.089	1.171	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Table 2B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 2

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	1503.514	706.598	0.470	1.504	1.014	Inf
2	1583.205	648.158	0.409	1.583	1.142	Inf
3	1638.104	544.559	0.332	1.638	1.275	Inf
4	1412.134	941.423	0.667	1.412	0.715	Inf
5	1503.514	706.598	0.470	1.504	1.014	Inf
6	134.590	-1089.118	-8.092	0.135	0.529	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Section 3: x = 24 in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 18240.000

Design Geometry

Width h (in) = 5.896

Depth, b (in) = 3.959

Flange thickness tf (in) = 0.196

Web thickness tw (in) = 0.257

Failure Predictions

Critical stress - Crippling (psi) = 37949.266

Critical stress - Local Buckling (psi) =
86037.439

Analysis Information

Cross-sectional area (in²) = 2.965Moment of Inertia Iz (in⁴) = 16.177

Local Buckling Coefficient, kw = 4.000

(Limit) Max. Bending Stress (psi) = 3323.897

(Limit) Avg. Corner Stress (psi) = 3102.979

(Limit) Avg. Flange Stress (psi) = 3213.438

Factors of Safety

Ultimate Strength = 24.369

Yielding = 21.360

Crippling = 11.417

Local Buckling = 25.885

Table 3A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 3

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	-2141.290	-4581.730	2.140	-2.141	2.910	Inf
2	-1952.352	-4817.902	2.468	-1.952	2.970	Inf
3	-1653.202	-4985.846	3.016	-1.653	2.929	Inf
4	-2871.847	-4307.771	1.500	-2.872	2.872	Inf
5	-2141.290	-4581.730	2.140	-2.141	2.910	Inf
6	3306.403	-404.116	-0.122	3.306	3.551	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Table 3B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 3

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	4581.730	2141.290	0.467	4.582	3.100	Inf
2	4817.902	1952.352	0.405	4.818	3.491	Inf
3	4985.846	1653.202	0.332	4.986	3.884	Inf
4	4307.771	2871.847	0.667	4.308	2.180	Inf
5	4581.730	2141.290	0.467	4.582	3.100	Inf
6	404.116	-3306.403	-8.182	0.404	1.598	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Section 4: $x = 36$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 42120.000

Design Geometry

Width h (in) = 6.242Depth, b (in) = 4.183Flange thickness t_f (in) = 0.217Web thickness t_w (in) = 0.298

Failure Predictions

Critical stress - Crippling (psi) = 40261.535

Critical stress - Local Buckling (psi) =
104234.014

Analysis Information

Cross-sectional area (in²) = 3.545Moment of Inertia I_z (in⁴) = 21.326Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 6163.527

(Limit) Avg. Corner Stress (psi) = 5735.592

(Limit) Avg. Flange Stress (psi) = 5949.560

Factors of Safety

Ultimate Strength = 13.142

Yielding = 11.519

Crippling = 6.532

Local Buckling = 16.911

Table 4A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 4

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-3954.140	-8505.667	2.151	-3.954	5.396	Inf
2	-3584.328	-8932.373	2.492	-3.584	5.495	Inf
3	-3058.058	-9245.291	3.023	-3.058	5.428	Inf
4	-5336.666	-8004.999	1.500	-5.337	5.337	Inf
5	-3954.140	-8505.667	2.151	-3.954	5.396	Inf
6	6116.115	-739.623	-0.121	6.116	6.565	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Table 4B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 4

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	8505.667	3954.140	0.465	8.506	5.772	Inf
2	8932.373	3584.328	0.401	8.932	6.499	Inf
3	9245.291	3058.058	0.331	9.245	7.207	Inf
4	8004.999	5336.666	0.667	8.005	4.051	Inf
5	8505.667	3954.140	0.465	8.506	5.772	Inf
6	739.623	-6116.115	-8.269	0.740	2.942	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Section 5: x = 48 in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 76800.000

Design Geometry

Width h (in) = 6.528

Depth, b (in) = 4.374

Flange thickness tf (in) = 0.236

Web thickness tw (in) = 0.337

Failure Predictions

Critical stress - Crippling (psi) = 42288.611

Critical stress - Local Buckling (psi) =
122011.794

Analysis Information

Cross-sectional area (in²) = 4.101Moment of Inertia Iz (in⁴) = 26.654

Local Buckling Coefficient, kw = 4.000

(Limit) Max. Bending Stress (psi) = 9404.374

(Limit) Avg. Corner Stress (psi) = 8724.985

(Limit) Avg. Flange Stress (psi) = 9064.679

Factors of Safety

Ultimate Strength = 8.613

Yielding = 7.550

Crippling = 4.497

Local Buckling = 12.974

Table 5A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 5

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	-6009.395	-12992.143	2.162	-6.009	8.234	Inf
2	-5416.920	-13626.938	2.516	-5.417	8.365	Inf
3	-4655.165	-14106.561	3.030	-4.655	8.278	Inf
4	-8159.235	-12238.853	1.500	-8.159	8.159	Inf
5	-6009.395	-12992.143	2.162	-6.009	8.234	Inf
6	9310.330	-1114.418	-0.120	9.310	9.986	Inf

Damage = 0.000000e+00 Design Life = 4503599627370496.000 SF = 4.0 No. of Flights = Inf

Table 5B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 5

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	12992.143	6009.395	0.463	12.992	8.841	Inf
2	13626.938	5416.920	0.398	13.627	9.953	Inf
3	14106.561	4655.165	0.330	14.107	11.005	1561237346818.195
4	12238.853	8159.235	0.667	12.239	6.193	Inf
5	12992.143	6009.395	0.463	12.992	8.841	Inf
6	1114.418	-9310.330	-8.354	1.114	4.457	Inf

Damage = 3.202588e-18 Design Life = 450359e10 SF = 4.0 No. of Flights = 7806186e10

Section 6: x = 60 in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 123000.000

Design Geometry

Width h (in) = 6.777

Depth, b (in) = 4.546

Flange thickness tf (in) = 0.254

Web thickness tw (in) = 0.373

Failure Predictions

Critical stress - Crippling (psi) = 44117.264

Critical stress - Local Buckling (psi) =
139589.631

Analysis Information

Cross-sectional area (in²) = 4.645Moment of Inertia Iz (in⁴) = 32.220

Local Buckling Coefficient, kw = 4.000

(Limit) Max. Bending Stress (psi) = 12936.208

(Limit) Avg. Corner Stress (psi) = 11967.150

(Limit) Avg. Flange Stress (psi) = 12451.679

Factors of Safety

Ultimate Strength = 6.261

Yielding = 5.488

Crippling = 3.410

Local Buckling = 10.791

Table 6A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 6

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	-8235.001	-17889.829	2.172	-8.235	11.328	181453432357.963
2	-7383.104	-18741.726	2.538	-7.383	11.482	77624898364.088
3	-6389.225	-19404.312	3.037	-6.389	11.380	134730538265.648
4	-11245.036	-16867.553	1.500	-11.245	11.245	298019564247.972
5	-8235.001	-17889.829	2.172	-8.235	11.328	181453432357.963
6	12778.450	-1514.483	-0.119	12.778	13.697	66099015.844

Damage = 3.262764e-14 Design Life = 30648865e6 SF = 4.0 No. of Flights = 7662216e6

Table 6B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 6

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	17889.829	8235.001	0.460	17.890	12.205	3595005425.064
2	18741.726	7383.104	0.394	18.742	13.739	60560682.385
3	19404.312	6389.225	0.329	19.404	15.148	5116351.937
4	16867.553	11245.036	0.667	16.868	8.536	Inf
5	17889.829	8235.001	0.460	17.890	12.205	3595005425.064
6	1514.483	-12778.450	-8.438	1.514	6.091	Inf

Damage = 1.803433e-12 Design Life = 554497e6 SF = 4.0 No. of Flights = 138624e6

Section 7: $x = 72$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 181440.000

Design Geometry

Width h (in) = 7.002Depth, b (in) = 4.704Flange thickness t_f (in) = 0.271Web thickness t_w (in) = 0.407

Failure Predictions

Critical stress - Crippling (psi) = 45797.306

Critical stress - Local Buckling (psi) =
157086.601

Analysis Information

Cross-sectional area (in²) = 5.182Moment of Inertia I_z (in⁴) = 38.056Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 16691.602

(Limit) Avg. Corner Stress (psi) = 15398.738

(Limit) Avg. Flange Stress (psi) = 16045.170

Factors of Safety

Ultimate Strength = 4.853

Yielding = 4.254

Crippling = 2.744

Local Buckling = 9.411

Table 7A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 7

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-10587.245	-23105.946	2.182	-10.587	14.617	11867265.470
2	-9442.678	-24178.977	2.561	-9.443	14.783	9030045.230
3	-8226.575	-25037.403	3.043	-8.227	14.675	10775423.725
4	-14536.001	-21804.001	1.500	-14.536	14.536	13611843.099
5	-10587.245	-23105.946	2.182	-10.587	14.617	11867265.470
6	16453.150	-1931.457	-0.117	16.453	17.625	245559.669

Damage = 5.104700e-11 Design Life = 19589790224.943 SF = 4.0 No. of Flights = 4897447556.236

Table 7B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 7

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	23105.946	10587.245	0.458	23.106	15.802	2031237.204
2	24178.977	9442.678	0.391	24.179	17.787	208750.870
3	25037.403	8226.575	0.329	25.037	19.558	42824.103
4	21804.001	14536.001	0.667	21.804	11.034	1255288413120.147
5	23105.946	10587.245	0.458	23.106	15.802	2031237.204
6	1931.457	-16453.150	-8.519	1.931	7.809	Inf

Damage = 3.572617e-10 Design Life = 2799068905.226 SF = 4.0 No. of Flights = 699767226.307

Section 8: $x = 84$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 252840.000

Design Geometry

Width h (in) = 7.208Depth, b (in) = 4.851Flange thickness t_f (in) = 0.288Web thickness t_w (in) = 0.441

Failure Predictions

Critical stress - Crippling (psi) = 47360.545

Critical stress - Local Buckling (psi) =
174574.796

Analysis Information

Cross-sectional area (in²) = 5.717Moment of Inertia I_z (in⁴) = 44.179Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 20625.714

(Limit) Avg. Corner Stress (psi) = 18977.810

(Limit) Avg. Flange Stress (psi) = 19801.762

Factors of Safety

Ultimate Strength = 3.927

Yielding = 3.442

Crippling = 2.296

Local Buckling = 8.464

Table 8A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 8

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-13037.370	-28578.605	2.192	-13.037	18.064	159400.847
2	-11569.587	-29873.708	2.582	-11.570	18.230	136127.408
3	-10144.973	-30938.571	3.050	-10.145	18.125	150394.398
4	-17993.297	-26989.946	1.500	-17.993	17.993	170580.079
5	-13037.370	-28578.605	2.192	-13.037	18.064	159400.847
6	20289.946	-2359.965	-0.116	20.290	21.722	8840.964

Damage = 3.570490e-09 Design Life = 280073591.028 SF = 4.0 No. of Flights = 70018397.757

Table 8B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 8

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	28578.605	13037.370	0.456	28.579	19.589	41766.043
2	29873.708	11569.587	0.387	29.874	22.049	7148.755
3	30938.571	10144.973	0.328	30.939	24.183	2027.129
4	26989.946	17993.297	0.667	26.990	13.658	71801592.494
5	28578.605	13037.370	0.456	28.579	19.589	41766.043
6	2359.965	-20289.946	-8.598	2.360	9.591	Inf

Damage = 9.515616e-09 Design Life = 105090409.203 SF = 4.0 No. of Flights = 26272602.301

Section 9: $x = 96$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 337920.000

Design Geometry

Width h (in) = 7.399Depth, b (in) = 4.990Flange thickness t_f (in) = 0.304Web thickness t_w (in) = 0.474

Failure Predictions

Critical stress - Crippling (psi) = 48828.782

Critical stress - Local Buckling (psi) =
192101.022

Analysis Information

Cross-sectional area (in²) = 6.253Moment of Inertia I_z (in⁴) = 50.601Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 24706.944

(Limit) Avg. Corner Stress (psi) = 22674.951

(Limit) Avg. Flange Stress (psi) = 23690.948

Factors of Safety

Ultimate Strength = 3.278

Yielding = 2.874

Crippling = 1.976

Local Buckling = 7.775

Table 9A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 9

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-15565.375	-34264.039	2.201	-15.565	21.639	9340.726
2	-13746.045	-35780.147	2.603	-13.746	21.794	8436.268
3	-12128.863	-37060.416	3.056	-12.129	21.700	8972.325
4	-21589.377	-32384.065	1.500	-21.589	21.589	9656.406
5	-15565.375	-34264.039	2.201	-15.565	21.639	9340.726
6	24257.727	-2796.377	-0.115	24.258	25.955	815.690

Damage = 6.092919e-08 Design Life = 16412494.812 SF = 4.0 No. of Flights = 4103123.703

Table 9B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 9

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	34264.039	15565.375	0.454	34.264	23.538	2905.332
2	35780.147	13746.045	0.384	35.780	26.491	631.861
3	37060.416	12128.863	0.327	37.060	28.985	212.774
4	32384.065	21589.377	0.667	32.384	16.388	965434.689
5	34264.039	15565.375	0.454	34.264	23.538	2905.332
6	2796.377	-24257.727	-8.675	2.796	11.421	107465402189.605

Damage = 1.038367e-07 Design Life = 9630505.449 SF = 4.0 No. of Flights = 2407626.362

Section 10: $x = 108$ in — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 437400.000

Design Geometry

Width h (in) = 7.579Depth, b (in) = 5.123Flange thickness t_f (in) = 0.320Web thickness t_w (in) = 0.506

Failure Predictions

Critical stress - Crippling (psi) = 50217.736

Critical stress - Local Buckling (psi) =
209697.177

Analysis Information

Cross-sectional area (in²) = 6.789Moment of Inertia I_z (in⁴) = 57.334Local Buckling Coefficient, k_w = 4.000

(Limit) Max. Bending Stress (psi) = 28912.039

(Limit) Avg. Corner Stress (psi) = 26468.623

(Limit) Avg. Flange Stress (psi) = 27690.331

Factors of Safety

Ultimate Strength = 2.802

Yielding = 2.456

Crippling = 1.737

Local Buckling = 7.253

Table 10A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 10

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	-18156.760	-40129.910	2.210	-18.157	25.324	1114.688
2	-15959.445	-41864.632	2.623	-15.959	25.454	1044.196
3	-14166.899	-43368.058	3.061	-14.167	25.381	1082.837
4	-25303.816	-37955.724	1.500	-25.304	25.304	1125.895
5	-18156.760	-40129.910	2.210	-18.157	25.324	1114.688
6	28333.798	-3238.148	-0.114	28.334	30.300	126.781

Damage = 5.141617e-07 Design Life = 1944913.456 SF = 4.0 No. of Flights = 486228.364

Table 10B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 10

Load Case	$\sigma_{\max}$ (psi)	$\sigma_{\min}$ (psi)	R	$S_{\max}$ (ksi)	S_{eq} (ksi)	$N_f \times 10^6$
1	40129.910	18156.760	0.452	40.130	27.624	378.057
2	41864.632	15959.445	0.381	41.865	31.089	94.429
3	43368.058	14166.899	0.327	43.368	33.937	35.465
4	37955.724	25303.816	0.667	37.956	19.207	57190.687
5	40129.910	18156.760	0.452	40.130	27.624	378.057
6	3238.148	-28333.798	-8.750	3.238	13.289	163490665.731

Damage = 6.845166e-07 Design Life = 1460885.008 SF = 4.0 No. of Flights = 365221.252

Section 11: x = 120 in (fuselage) — Crippling drives design

Requirements

(Limit) Bending Moment (in-lb) = 552000.000

Design Geometry

Width h (in) = 7.750

Depth, b (in) = 5.250

Flange thickness tf (in) = 0.336

Web thickness tw (in) = 0.537

Failure Predictions

Critical stress - Crippling (psi) = 51539.189

Critical stress - Local Buckling (psi) =
227385.768

Analysis Information

Cross-sectional area (in²) = 7.329Moment of Inertia Iz (in⁴) = 64.383

Local Buckling Coefficient, kw = 4.000

(Limit) Max. Bending Stress (psi) = 33223.300

(Limit) Avg. Corner Stress (psi) = 30342.518

(Limit) Avg. Flange Stress (psi) = 31782.909

Factors of Safety

Ultimate Strength = 2.438

Yielding = 2.137

Crippling = 1.551

Local Buckling = 6.844

Table 11A. Wing Spar Fatigue (Design Loads) - TOP SURFACE - Section 11

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	-20800.675	-46151.497	2.219	-20.801	29.101	202.964
2	-18200.590	-48101.560	2.643	-18.201	29.195	195.452
3	-16250.527	-49834.950	3.067	-16.251	29.153	198.787
4	-29120.945	-43681.417	1.500	-29.121	29.121	201.341
5	-20800.675	-46151.497	2.219	-20.801	29.101	202.964
6	32501.054	-3683.453	-0.113	32.501	34.738	27.497

Damage = 2.846904e-06 Design Life = 351258.763 SF = 4.0 No. of Flights = 87814.691

Table 11B. Wing Spar Fatigue (Design Loads) - BOTTOM SURFACE - Section 11

Load Case	sigma_max (psi)	sigma_min (psi)	R	Smax (ksi)	Seq (ksi)	Nf x10 ⁶
1	46151.497	20800.675	0.451	46.151	31.832	72.242
2	48101.560	18200.590	0.378	48.102	35.822	19.741
3	49834.950	16250.527	0.326	49.835	39.018	8.008
4	43681.417	29120.945	0.667	43.681	22.105	6899.361
5	46151.497	20800.675	0.451	46.151	31.832	72.242
6	3683.453	-32501.054	-8.824	3.683	15.187	4829991.740

Damage = 3.257318e-06 Design Life = 307001.027 SF = 4.0 No. of Flights = 76750.257

4. WEIGHT OF A SINGLE WING SPAR

Weight Calculation Method:

The cross-sectional area $A(x)$ was first computed at the 11 required spanwise stations. A fourth-order polynomial was then fit to the discrete $A(x)$ data using polyfit. The resulting polynomial was integrated analytically from $x = 0$ to $x =$ using polyint to obtain the spar volume. Finally, the spar weight was calculated from $W = \rho * V$ using the assumed material density.

Supporting Weight Calculations:

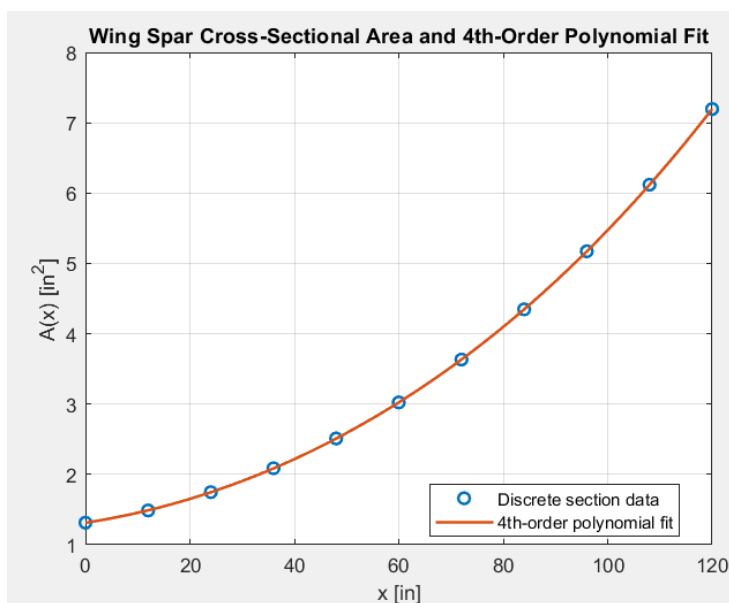
Material density, $\rho = 0.102000 \text{ lb/in}^3$

Span length, $L = 120.000 \text{ in}$

$x \text{ (in)}$	$A(x) \text{ [in}^2\text{]}$	$A_{\text{fit}}(x) \text{ [in}^2\text{]}$
120.0	7.329	7.314
108.0	6.789	6.814
96.0	6.253	6.264
84.0	5.717	5.704
72.0	5.182	5.158
60.0	4.645	4.632
48.0	4.101	4.115
36.0	3.545	3.581
24.0	2.965	2.984
12.0	2.329	2.263
0.0	1.314	1.341

Estimated spar volume = 550.705 in^3

Estimated spar weight = 56.172 lb



5. TIP DEFLECTION

Deflection Integration Method:

The tip deflection was computed using Euler-Bernoulli beam theory with curvature defined by $\kappa(x) = M(x)/(E I(x))$. The spanwise moment of inertia $I(x)$ was obtained from the 11 discrete design stations and interpolated along the beam using a linear interpolation. The deflection calculation used the design-load bending moment distribution, $M_{\text{design}}(x) = 1.5 M(x)$. The curvature was then integrated twice numerically using `cumtrapz`, applying cantilever boundary conditions of zero slope and zero deflection at the root.

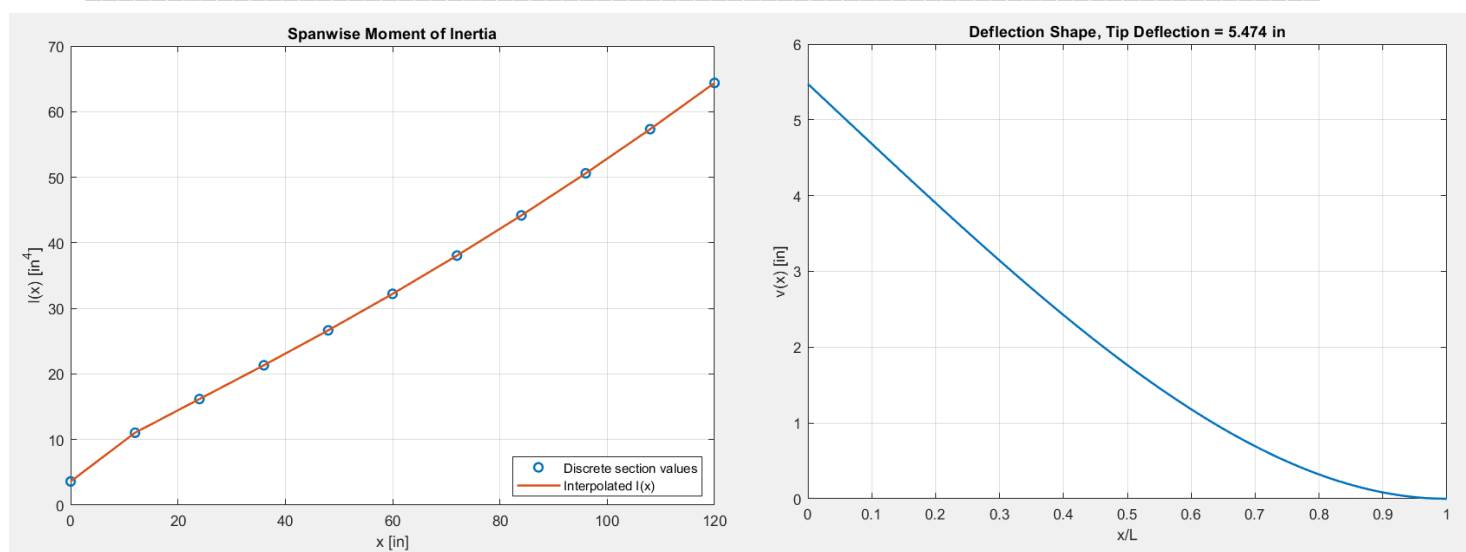
Supporting Deflection Calculations:

Elastic modulus, $E = 1.070\text{e}+07$ psi

Design-load scaling factor on moment = 1.500

Maximum design-load root bending moment = 828.000 kip-in

Tip deflection = 5.474 in



6. GOVERNING FACTOR OF SAFETY SUMMARY

Section	x [in]	Driving Failure Mode	Governing FS
1	0	None (no loading)	∞
2	12	Crippling	32.206
3	24	Crippling	11.417
4	36	Crippling	6.532
5	48	Crippling	4.497
6	60	Crippling	3.410
7	72	Crippling	2.744
8	84	Crippling	2.296
9	96	Crippling	1.976
10	108	Crippling	1.737
11	120	Crippling	1.551

Note: Section 1 experiences zero bending moment; therefore, all stresses and factors of safety are theoretically infinite.

